

# weatherisk: A Python Package for Spatial Clustering of Climate Extremes via Max-Stable Processes

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## Abstract

We present **weatherisk**, an open-source Python package for identifying regions of homogeneous extremal dependence in gridded climate-model output. The package implements a complete analysis pipeline: seasonal-trend decomposition (STL) removes long-term trends from raw monthly fields; annual block maxima are extracted and fitted with generalised extreme-value (GEV) distributions; marginals are transformed to unit Fréchet scale—the standard heavy-tailed reference form required by max-stable models—and local spatial-dependence parameters are estimated before grid cells are partitioned into coherent regions. Two clustering schemes are supported—Local Estimates Clustering (LEC) and Extremal Dependence Clustering (EDC)—each validated at every pipeline stage against an established R reference implementation to ensure numerical fidelity. We demonstrate **weatherisk** on a global-precipitation case study using the AWI-ESM-1-1-LR model (T63 grid,  $192 \times 96$  cells, 1850–2005), executed on the AWI *albedo* high-performance cluster, reproducing a published cluster map entirely in Python. The result is a reproducible, scalable framework for spatial-extremes analysis, suited equally to synthetic benchmarks and publication-scale climate-model output.

*Keywords:* spatial extremes, max-stable process, spatial clustering, extremal dependence, Python

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## Metadata

Table 1 summarises the mandatory software metadata for the current release of **weatherisk**.

| Nr. | Code metadata description                                       | Metadata                                                                                                                                                           |
|-----|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1  | Current code version                                            | v0.1.0                                                                                                                                                             |
| C2  | Permanent link to code/repository                               | <a href="https://github.com/wkchikach99-byte/weatherisk">https://github.com/wkchikach99-byte/weatherisk</a>                                                        |
| C3  | Permanent link to Reproducible Capsule                          | <a href="https://doi.org/10.5281/zenodo.20069739">https://doi.org/10.5281/zenodo.20069739</a>                                                                      |
| C4  | Legal Code License                                              | MIT                                                                                                                                                                |
| C5  | Code versioning system used                                     | git                                                                                                                                                                |
| C6  | Software code languages, tools, and services used               | Python 3.10+; optional Rust backend; R (validation); SLURM                                                                                                         |
| C7  | Compilation requirements, operating environments & dependencies | Python $\geq 3.10$ ; NumPy; SciPy; pandas; matplotlib; Click; xarray; netCDF4; scikit-learn; Cartopy; h5py; statsmodels; optional pytest/ruff/mypy for development |
| C8  | Link to developer documentation                                 | <a href="https://github.com/wkchikach99-byte/weatherisk">https://github.com/wkchikach99-byte/weatherisk</a>                                                        |
| C9  | Support email for questions                                     | widad.kchikach@awi.de                                                                                                                                              |

Table 1: Code metadata (mandatory).

## 1. Motivation and significance

Extreme weather events such as heavy precipitation or heat waves rarely act on a single location; they typically unfold across broad regions at once [1]. Understanding *where* such events tend to co-occur matters enormously for regional risk assessment, reinsurance, and infrastructure planning. Ignoring the spatial footprint of extremes can lead to a serious underestimate of aggregate losses.

While mean climate trends carry important information, socioeconomic impacts are driven disproportionately by extreme events rather than by shifts in average conditions [1]. Infrastructure failures, crop losses, and flood damage are governed by tail probabilities (the chances of exceeding a rarely reached, very high threshold)—not by means—of precipitation and temperature distributions. Getting the tail behaviour and its spatial dependence structure right is therefore central to any serious risk quantification.

Extremes also behave differently from everyday climate variability: they show non-Gaussian distributions, heavy tails (where extreme values occur far more often than a bell-curve distribution would predict), and asymptotic dependence—also called *extremal dependence* throughout this paper—none of which standard covariance-based techniques adequately represent [2]. Extreme value theory (EVT) provides the rigorous mathematical framework for this regime. A central concept in EVT is the *block maximum*: the single largest observed value within a fixed time window—here, one calendar year. Annual block maxima follow a *generalised extreme-value* (GEV) distribution, a three-parameter family that characterises how the magnitude of the largest events grows with record length. Crucially, this result holds regardless of the underlying data distribution, which is why EVT methods remain valid even over the centuries-long simulation periods typical of climate modelling [2,3]. Capturing these complex spatial patterns requires both a statistically sound framework and software that can handle the associated computational burden—the dual motivation for the present work.

*Max-stable processes* are the standard framework for modelling joint tail behaviour across space [2,3]. Following Contzen et al. [4], fitting these models at each location yields anisotropy parameters that capture the strength and directional orientation of local spatial dependence. Grid cells with similar parameter estimates can then be grouped into *clusters*—geographically connected regions in which the extremal dependence structure is approximately uniform.

An R-based pipeline that implements this workflow was developed by Contzen et al. [4]. It simulates max-stable processes, estimates local anisotropy parameters via pairwise composite likelihood [3] (see Section 2.1 for the technical details), computes a pairwise dissimilarity score between all grid cells based on the geometric overlap of their fitted anisotropy ellipses, and partitions cells by hierarchical clustering. In practice, however, the implementation is fragmented across independent R scripts tied together by SLURM shell files, which makes installation, testing, and extension unnecessarily difficult. This limitation becomes particularly acute in climate-model applications, where extremes are, by definition, rare, and stable statistical inference requires long time series—decades or centuries. Modern coupled models such as AWI-ESM-1-1-LR produce monthly output on global grids (e.g. T63,  $192 \times 96 = 18,432$  cells) spanning hundreds of simulated years. Processing these heavy NetCDF archives before any clustering can begin involves detrending, block-maxima extraction, GEV marginal fitting (estimating a separate GEV distribution independently at each grid cell), and Fréchet transformation at every grid cell. Scientific software for this problem therefore has to do more than implement the mathematics: it must also

provide a reproducible preprocessing chain, checkpoint/restart support for long-running jobs, and numerically efficient kernels that scale from laptops to high-performance computing facilities.

**weatherisk** solves this problem by re-implementing the full pipeline in a single Python package that can be installed with `pip install`. It faithfully reproduces every step of the original R code at reduced and controlled scales and adds several practical features:

- a complete preprocessing chain for climate-model output: STL (Seasonal-Trend decomposition using Loess) detrending, annual block-maxima extraction, GEV fitting, and Fréchet transformation (a probability-integral step that maps each grid cell’s fitted GEV values to a common unit-Fréchet scale required by the max-stable model);
- checkpoint/restart support and scalable execution strategies for heavy CMIP6 (Coupled Model Intercomparison Project Phase 6)-style workloads on high-performance computing (HPC) infrastructure; and
- a 196-test validation suite, including 30 end-to-end CMIP6 parity tests (Section 4).

To demonstrate the software on a publication-level workload, we include a Python-based revisit of the Figure 9 case study from Contzen et al. [4] on the AWI-ESM-1-1-LR historical precipitation dataset and provide a systematic model-validation analysis (Section 4) documenting the precision of every pipeline stage against the original R implementation.

Existing Python packages do support parts of the extreme-value workflow. For example, **pyextremes** [5] and **scikit-extremes**<sup>1</sup> focus on univariate extreme-value analysis, while the R package **SpatialExtremes** [6] provides max-stable simulation. However, to our knowledge, no existing Python package offers the integrated spatial-extremes workflow implemented here: max-stable simulation, local composite-likelihood estimation, and ellipse-based spatial clustering of extremes in one package.

## 2. Software description

### 2.1. Software architecture

**weatherisk** is a pip-installable Python package organised into 18 modules and a CLI entry point. Figure 1 shows the overall workflow.

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<sup>1</sup><https://github.com/kikocorreoso/scikit-extremes>

The processing pipeline consists of seven stages. The first stage contains three sub-steps that together convert raw climate-model output into the standardised representation required by the subsequent statistical analysis:

1. **Preprocessing.**

- (a) **STL decomposition (detrending).** Raw monthly precipitation fields are decomposed using the STL method of Cleveland et al. [10] with `s.window = ``periodic''` and `robust = TRUE`. The trend component is subtracted, retaining the seasonal cycle and residuals. This step isolates the stationary signal needed for extreme-value modelling. Both a native Python implementation and an optional R-subprocess path (for exact cross-validation against the original R workflow) are provided.

- (b) **Annual block-maxima extraction.** From the detrended monthly series, the annual maximum precipitation value is extracted for each grid cell, keeping only years with all 12 months present. The result is an  $(n_{\text{years}} \times n_{\text{cells}})$  matrix of annual extremes that feeds the marginal fitting step. Taking the annual maximum is the standard “block maxima” approach of classical EVT [2].

- (c) **GEV fitting and Fréchet transformation.** A GEV distribution is fitted independently at each grid cell by maximum likelihood. Before fitting, cells that are entirely missing or have near-zero variance are excluded; isolated missing values in the remaining cells are replaced by the cell-wise mean. Any cell that still yields non-finite GEV or Fréchet values after fitting is dropped. In the full AWI-ESM-1-1-LR run, all  $192 \times 96 = 18,432$  cells passed this screen. The fitted GEV at each cell is then used to apply the probability integral transform: each annual maximum is mapped through its own GEV cumulative distribution function, producing a value on the unit Fréchet scale (the common marginal form required by max-stable models, as described in Section 1). This puts all cells on an equal footing before spatial dependence is estimated: what remains captures where extremes co-occur, not how large they are at any individual site. Values are clipped to  $[0.05, n_{\text{years}}^2]$  to suppress numerical artefacts at the tails.

2. **Simulation** (for validation on synthetic data). Synthetic max-stable random fields with user-specified spatially varying anisotropy are generated on a regular grid following the spectral construction of Schlather [7]. These fields serve as controlled test cases whose true parameters are known by construction.

3. **Local estimation.** For each grid cell, we estimate the three local anisotropy parameters—semi-axis lengths  $a$  and  $b$  (controlling the

range and degree of directional asymmetry of spatial dependence) and a rotation angle  $\gamma$  (the principal direction of dependence)—by maximising a pairwise composite likelihood [3,4], which sums log-likelihood contributions over neighbouring grid-cell pairs; this avoids fitting the full joint spatial distribution at once and makes estimation computationally feasible on large grids. The optimisation runs over all neighbouring pairs within grid-point distance  $\varepsilon$  (default  $\varepsilon = 5$ ). Neighbours are enumerated in a fixed stencil order (x-offset varies fastest) matching the R reference, and at least three valid neighbours are required; cells with fewer than five valid pairs receive default estimates  $(a, b, \gamma) = (1, 0, 0)$ . A multi-start strategy with five random restarts (seed = 42) mitigates sensitivity to local optima [4]. Fréchet values are normalised to 11 decimal places before optimisation to ensure numerical parity with the R workflow at the precision boundary.

4. **Spatial smoothing.** The raw estimates  $(a_i, b_i, \gamma_i)$  are smoothed by a moving average over grid cells within a smoothing radius (default 2.0 in grid-point units). Angular wrapping is applied to the rotation angle  $\gamma$  to account for its directional periodicity ( $+90^\circ \equiv -90^\circ$ ).
5. **Clustering.** From the smoothed parameter estimates, we assemble a pairwise dissimilarity matrix and apply average-linkage hierarchical agglomerative clustering [4] to group grid cells into regions. The number of clusters  $k$  is set automatically by applying a quantile threshold (default: 30<sup>th</sup> percentile) to the dendrogram merge heights, following Contzen et al. [4]. Two dissimilarity measures are available:
  - **LEC** (Local Estimates Clustering [4]; ellipse-overlap dissimilarity  $D_2$ ): Jaccard-like overlap of the anisotropy ellipses (a ratio of intersection to union area, ranging from 0 for non-overlapping to 1 for identical ellipses; here used as a geometrically interpretable dissimilarity  $1 - \text{overlap}$ ). Each grid cell’s fitted parameters  $(a, b, \gamma)$  define an ellipse in 2D space. Two cells whose ellipses overlap strongly have a similar dependence geometry, so high overlap means low dissimilarity. The ratio  $D_2$  was introduced by Contzen et al. [4] as a geometrically clear way to measure this.
  - **EDC** (Extremal Dependence Clustering [4]; madogram-based dissimilarity  $D_1$ , following Cooley et al. [9] and Saunders et al. [8]): The madogram is a rank-based summary of extremal dependence: computed from order statistics without parametric assumptions, it maps onto pairwise extremal coefficients that range from 1 (complete dependence: extreme events at the two sites always co-occur) to 2 (independence), with intermediate values reflecting partial co-

occurrence [9]. Contzen et al. [4] adopt this approach alongside the ellipse method, providing an alternative non-parametric dissimilarity  $D_1$  for dependence-based regionalisation.

6. **In-cluster re-estimation.** Within each cluster, the dependence parameters are re-estimated using all data from the cluster’s grid cells via global pairwise composite-likelihood maximum-likelihood estimation (MLE; distance cutoff:  $4.0 \times$  neighbour radius, 3 restarts per cluster). The goodness-of-fit of each method is assessed.

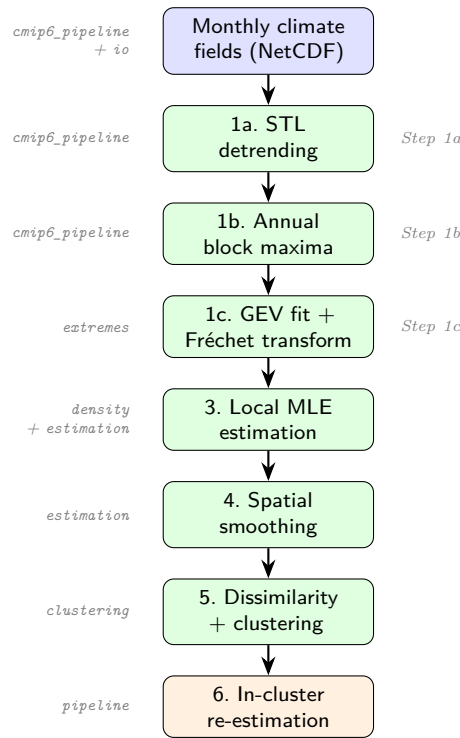


Figure 1: Processing pipeline of **weatherisk**. Stage 1 (a–c) is the preprocessing chain added for climate-model data; stages 3–6 implement the core statistical methodology of Contzen et al. [4]. Module names are shown on the left.

## 2.2. Software functionalities

Section 2.1 describes the end-to-end workflow. This subsection instead summarises the main software components and their roles within the package.

*Covariance modelling (covariance).* This module implements the anisotropic covariance functions used by the max-stable dependence model, including stationary and non-stationary formulations and the conversion between covariance and extremal-coefficient representations.

*Simulation (**simulation**)*. The simulation module generates synthetic max-stable fields with spatially varying anisotropy. These synthetic datasets are used for method development, controlled validation experiments, and regression tests against known parameter fields.

*Preprocessing and extremes (**cmip6\_pipeline**, **netcdf**, **extremes**)*. These components handle gridded climate-model input, including NetCDF loading, STL detrending of monthly series, annual block-maxima extraction, GEV fitting, and transformation to unit Fréchet margins. Together they provide the climate-data preprocessing chain required before spatial dependence can be estimated.

*Estimation (**density**, **estimation**)*. These modules implement pairwise composite-likelihood estimation of the local anisotropy parameters  $(a, b, \gamma)$ , the subsequent spatial smoothing step, and the in-cluster re-estimation routines used after clustering. They also contain the numerical safeguards needed for stable optimisation on large grids.

*Clustering (**clustering**)*. This module provides the two regionalisation methods implemented in **weatherisk**: LEC (Local Estimates Clustering), based on ellipse-overlap dissimilarity, and EDC (Extremal Dependence Clustering), based on madogram-derived extremal dependence. It also provides the hierarchical clustering utilities that convert these dissimilarities into region labels.

*Scalability and execution (**scalable**, **cli**)*. These components support large production runs through parallel execution, checkpoint/restart handling, and command-line entry points for the main workflows. They are designed to make the same code base usable both for local experiments and for HPC-scale Figure 9 case-study analysis.

*Plotting and outputs (**plotting**, **map\_plotting**)*. The plotting modules generate diagnostic figures, synthetic-study visualisations, and the publication-style cluster maps used in the CPC and CMIP6 workflows. They provide the final user-facing outputs of the pipeline in addition to the saved numerical arrays.

### 2.3. Computational performance

The dominant costs are the  $O(n^2)$  pairwise dissimilarity matrix and per-cell local MLE optimisation. **weatherisk** reduces both in four ways:

1. **Batch covariance.** All  $n^2$  entries are evaluated in a single vectorised NumPy call;  $2 \times 2$  matrix inverses use the analytic closed-form formula.



2. **Pre-drawn ellipse overlap (LEC).** Each fitted ellipse is rasterised once; pairwise overlap areas are computed with bitwise operations over batches of 256 pairs.
3. **Fast pairwise distance (EDC).** `scipy.spatial.distance.cdist` evaluates all pairwise rank-based distances in a single compiled-C call.
4. **Parallel local estimation.** Per-cell MLE tasks are distributed via Python’s `multiprocessing`; shared data is loaded once per worker.

On the same  $6 \times 6$  synthetic input, the Python pipeline completed in 17.9s compared with 43.9s for the original R implementation—a  $2.5\times$  speed-up on an identical workload. The full AWI-ESM-1-1-LR T63 grid (18,432 cells, 156 years) was run on the AWI *albedo* HPC cluster using 16 parallel workers, completing in approximately 4 hours with peak memory below 2 GiB; SLURM scripts and setup are provided under `hpc/`.

*Command-line interface (cli).* A single entry point exposes the main pipeline flows:

```
$ weatherisk validate --params stripes \
    --resolution 10 --n-sim 20 --seed 42
$ weatherisk maps --variable precip
```

### 3. Illustrative examples

We demonstrate `weatherisk` with two examples: one on synthetic data and one Python-based revisit of Figure 9 from Contzen et al. [4] on the AWI-ESM-1-1-LR climate model.

#### 3.1. Example 1: Synthetic validation

In this example, we simulate max-stable data on a  $51 \times 51$  grid where the semi-major axis  $b$  increases linearly from left to right, creating vertical “stripe” regions. The pipeline should recover these stripes from the simulated data alone.

```
from weatherisk.parameters import get_preset
from weatherisk.pipeline import run_pipeline

result = run_pipeline(
    resolution=51, n_sim=20,
    preset="stripes", seed=42,
)
```

Listing 1: Running the stripes validation.

Figure 2 shows the result. Panel (a) displays the EDC clusters ( $k_{\text{edc}} = 7$ ), panel (b) the LEC clusters ( $k_{\text{lec}} = 5$ ), and panel (c) the true parameter field. The true semi-major axis  $b$  increases smoothly from left to right (panel c), producing a continuous gradient from dark blue ( $b \approx 0$ ) on the left to light shading ( $b \approx 5$ ) on the right. The EDC method (panel a) uses seven clusters and captures the overall left-to-right trend, but its boundaries are irregular: some clusters extend horizontally across the domain, and the transitions between adjacent regions do not follow the vertical orientation of the true gradient. The LEC method (panel b), by contrast, uses only five clusters yet produces boundaries that are predominantly vertical, closely tracking the true transitions in the parameter field. The estimated within-cluster  $b$  values in the LEC map also progress monotonically from dark to light, mirroring the ground truth more faithfully than the EDC partition. This controlled experiment illustrates that the Python implementation correctly recovers the known spatial structure, consistent with the methodology of Contzen et al. [4]. Formal verification of numerical equivalence with the original R code is provided by the automated cross-validation tests described in Section 4.

Fig. 3 — EDC vs LEC ( $k_{\text{edc}}=7$ ,  $k_{\text{lec}}=5$ )

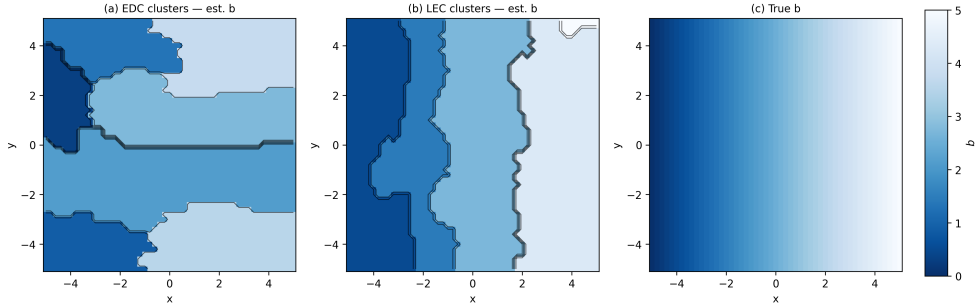


Figure 2: Synthetic validation on the “stripes” preset ( $k_{\text{edc}} = 7$ ,  $k_{\text{lec}} = 5$ ). (a) EDC clusters coloured by estimated semi-major axis  $b$ ; (b) LEC clusters coloured by estimated  $b$ ; (c) true parameter field showing a smooth left-to-right gradient in  $b$  from 0 to 5. The LEC method recovers sharper, vertically aligned boundaries that closely match the ground truth, while the EDC method produces more irregular, horizontally extended partitions.

### 3.2. Example 2: Revisiting the Figure 9 case study — AWI-ESM-1-1-LR global precipitation

To demonstrate the software on the same data used in the original publication, we revisit the Figure 9 analysis of Contzen et al. [4]. The experiment uses monthly precipitation from the AWI-ESM-1-1-LR historical simulation

| Parameter                      | Value                       |
|--------------------------------|-----------------------------|
| Degrees of freedom $\nu$       | 5                           |
| Covariance exponent $\alpha$   | 1                           |
| Neighbour radius $\varepsilon$ | 5 (grid-point units)        |
| Clustering quantile threshold  | 30 <sup>th</sup> percentile |
| Model / experiment             | AWI-ESM-1-1-LR / historical |
| Grid / resolution              | T63 ( $192 \times 96$ )     |
| Period                         | 1850–2005 (156 years)       |

Table 2: Parameter settings used for the Figure 9 case-study analysis, following Section 5 of Contzen et al. [4]. The degrees of freedom  $\nu$  and covariance exponent  $\alpha$  are parameters of the Schlather max-stable dependence model [7]:  $\nu$  controls how quickly dependence decays with distance (higher  $\nu$  gives weaker long-range dependence), and  $\alpha \in (0, 2]$  controls the smoothness of the simulated fields.

(T63 grid,  $192 \times 96$  cells, r1i1p1f1, years 1850–2005) with the parameter settings listed in Table 2, following Section 5 of [4]:

The full seven-stage pipeline (Section 2.1) was executed on the AWI *albedo* HPC cluster (16 CPU cores; SLURM scripts in `hpc/`). The command-line invocation is:

```
$ python scripts/reproduce_fig9.py \
  --data-dir /path/to/AWI-ESM-1-1-LR/pr/ \
  --workers 16 --output-dir output/cmip6_fig9 \
  --dpi 300
```

Listing 2: Running the Figure 9 case study on HPC.

Figure 3 shows the result, and Table 3 compares the resulting cluster counts with those reported by Contzen et al. [4]:

| Method                   | Contzen et al. [4] | <code>weatherisk</code> |
|--------------------------|--------------------|-------------------------|
| EDC ( $k_{\text{edc}}$ ) | 104                | 142                     |
| LEC ( $k_{\text{lec}}$ ) | 24                 | 21                      |

Table 3: Cluster counts reported in Contzen et al. [4] and obtained with `weatherisk` for the full Figure 9 case-study run.

The LEC result ( $k_{\text{lec}} = 21$  vs. 24) is close; the remaining difference is consistent with the level of variation expected from differences in preprocessing conventions and from the flat optimisation landscape of the composite-likelihood surface at this scale. In particular, the Python workflow makes

explicit choices about detrending, GEV fitting, partial-missing-value handling, and post-fit removal of non-finite cells that are not described in the same operational detail in Contzen et al. [4]. The EDC discrepancy is larger ( $k_{\text{edc}} = 142$  vs. 104); we attribute this primarily to differences in preprocessing and missing- data handling—details that are not fully specified in the published paper —rather than to any algorithmic deviation. Our coding and testing work shows that, on controlled mini-pipeline inputs, the Python implementation reproduces the R reference cluster counts exactly for both methods (Section 4). This supports the correctness of the methodology itself. At the full Figure 9 scale, the most plausible explanation is therefore not that the method is wrong, but that the published workflow leaves some preprocessing choices implicit. A separate point concerns LEC only: we identified a logical indexing issue in the original R ellipse-dissimilarity code. More precisely, in the R function `calc_distance_ellipses` (`r_code/functions.R`, line 515 in the repository version used here), the empty-ellipse branch inside the nested loop over grid-cell indices `i` and `j` writes `dist_matrix[j] = 1` rather than `dist_matrix[i,j] <- 1`. Because `dist_matrix[j]` uses linear indexing in R, the routine still executes and returns an output map, but some LEC dissimilarity values can be placed in incorrect matrix positions. This issue may contribute to the small remaining LEC difference, but it does not explain the larger EDC difference, which is more naturally linked to preprocessing and valid-cell-mask differences.

**Figure 9 — Clustering of Precipitation Extremes**  
**AWI-ESM-1-1-LR historical (1850–2005)  $\nu=5.0$ ,  $\alpha=1.0$ ,  $\varepsilon=5.0$**

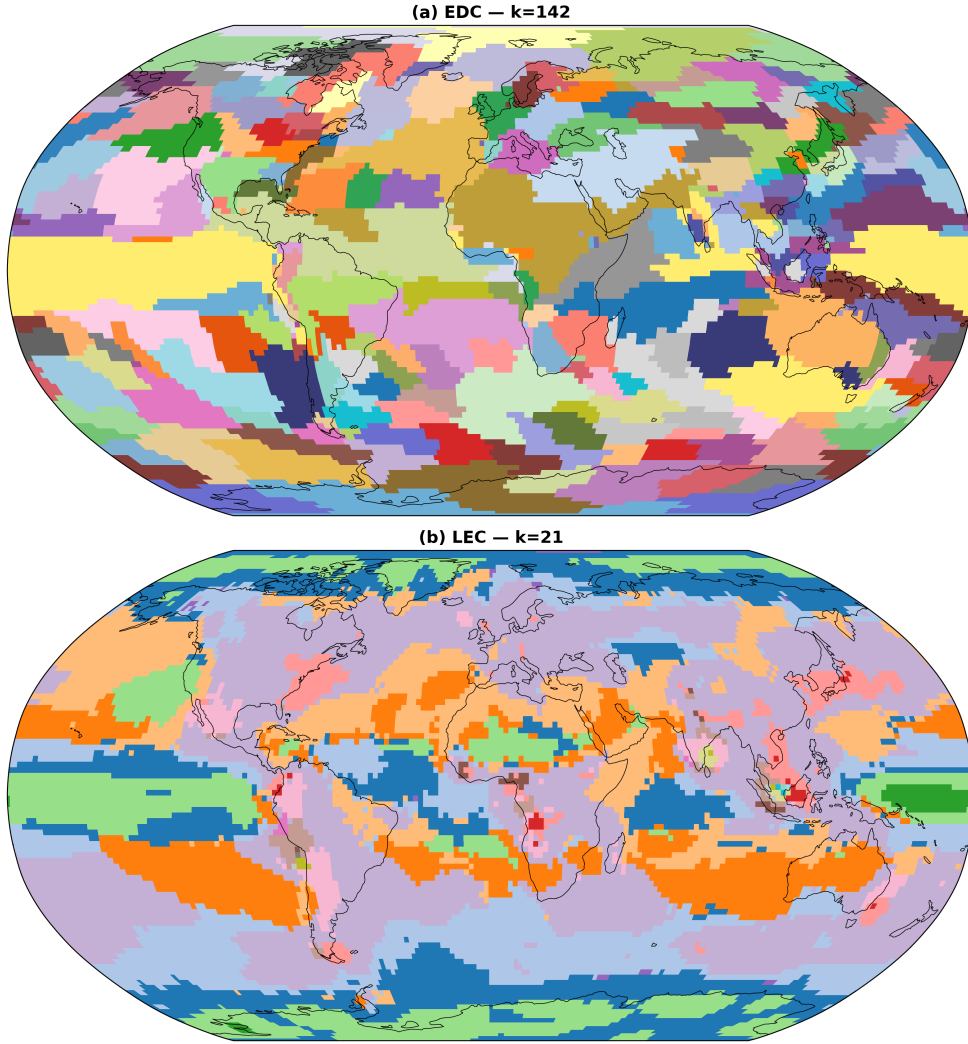


Figure 3: Python-based revisit of Figure 9 from Contzen et al. [4] using `weatherisk`. Left: EDC clusters ( $k_{\text{edc}} = 142$ ). Right: LEC clusters ( $k_{\text{lec}} = 21$ ). The experiment uses AWI-ESM-1-1-LR historical precipitation on the T63 grid (1850–2005) with  $\nu = 5$ ,  $\alpha = 1$ ,  $\varepsilon = 5$ , and 30% quantile threshold. The LEC panel shows spatially coherent regions consistent with the published Figure 9; the EDC panel captures the expected global heterogeneity of extremal dependence.

#### 4. Model validation

We verify `weatherisk` against the original R implementation of Contzen et al. [4] at three complementary levels (Table 5). **Level 1 (40 tests):** each mathematical kernel is tested against R reference values; all pass with relative errors  $\leq 10^{-10}$ , after resolving cross-language differences in matrix storage order (Fortran vs. C), angular periodicity of rotation angles, and likelihood normalisation conventions. **Level 2 (30 tests):** a mini AWI-ESM-1-1-LR subset ( $5 \times 5$  grid, 20 years) is processed through the full seven-stage pipeline in both Python and R; Table 4 lists per-stage agreement, and both EDC and LEC produce identical cluster counts on this controlled input. At full scale, counts differ from Contzen et al. ( $k_{\text{lec}} = 21$  vs. 24;  $k_{\text{edc}} = 142$  vs. 104), which we attribute to undocumented preprocessing choices and a known linear-index assignment in the original R ellipse-dissimilarity routine (`functions.R`, line 515). **Level 3 ( $\sim 126$  tests):** simulation distributional properties, checkpoint-resume, serial-parallel equivalence, and parameter-preset reproducibility.

| Pipeline stage                    | Tests     | Agreement criterion               |
|-----------------------------------|-----------|-----------------------------------|
| Detrending (STL)                  | 2         | mean absolute difference $< 0.01$ |
| Annual block-maxima extraction    | 1         | relative error $< 10^{-10}$       |
| GEV fitting and Fréchet transform | 1         | relative error $< 0.002$          |
| Local parameter estimation        | 5         | relative error $< 10^{-6}$        |
| Spatial smoothing                 | 2         | relative error $< 10^{-10}$       |
| EDC dissimilarity matrix          | 3         | relative error $< 10^{-10}$       |
| EDC cluster count and labels      | 4         | exact                             |
| LEC cluster count and labels      | 4         | exact                             |
| Python–Rust backend equivalence   | 3         | identical output                  |
| Pipeline configuration defaults   | 5         | exact                             |
| <b>Total</b>                      | <b>30</b> |                                   |

Table 4: Per-stage agreement between the Python and R implementations on a mini AWI-ESM-1-1-LR subset ( $5 \times 5$  grid, 20 years).

| Validation layer                 | Tests      | Agreement criterion            |
|----------------------------------|------------|--------------------------------|
| Per-function R parity            | 40         | relative error $\leq 10^{-10}$ |
| CMIP6 end-to-end parity          | 30         | exact cluster count            |
| Serial-parallel equivalence      | 5          | identical output               |
| Checkpoint integrity             | 4          | correct resume                 |
| Parameter-preset reproducibility | 30         | exact                          |
| Component & regression           | $\sim 87$  | per-test criteria              |
| <b>Total</b>                     | <b>196</b> |                                |

Table 5: Summary of the automated validation suite. All 196 tests pass in under 5 minutes on a standard laptop. The CMIP6 parity tests verify exact cluster-count agreement on a mini AWI-ESM-1-1-LR subset.

## 5. Impact

The core output of **weatherisk** is a small number of spatially coherent regions, each characterised by a common extremal dependence structure. Working with these regions rather than thousands of individual grid cells makes the analysis tractable and raises concrete applied questions: which areas share a common extreme-precipitation hazard, how do hazard boundaries shift across climate states, and where should spatial risk aggregation be drawn in insurance or infrastructure planning?

**weatherisk** makes this methodology directly accessible to any researcher working in Python. It packages a previously fragmented R/Shell/SLURM workflow as a standard, pip-installable library that requires no manual dependency management beyond a single `pip install` command.

*Reproducibility and correctness..* The validation suite described in Section 4 provides three layers of assurance: (i) per-function agreement with the R reference to  $\leq 10^{-10}$  relative error, (ii) exact cluster-count reproduction on a controlled CMIP6 subset, and (iii) structural tests covering checkpointing, parallelism, and parameter integrity. All 196 tests pass on every commit via continuous integration. A dedicated smoke test script (`scripts/release_smoke_test.sh`) provides a quick end-to-end check: it installs the package in a clean environment, runs the CLI, and confirms that the core pipeline completes correctly within about one minute.

*Computational efficiency..* The vectorisation and parallelisation strategies described in Section 2.3 yield large practical speed-ups. On a  $21 \times 21$  synthetic grid (441 cells), the EDC dissimilarity matrix is computed in 0.02 s, and the

full pipeline including 8-core parallel estimation completes in under 1 s. On the full AWI-ESM-1-1-LR T63 grid (18,432 cells, 156 years), the pipeline completed in approximately 4 hours on 16 cores of the AWI *albedo* high-performance computing HPC cluster.

That quadratic cost remains the fundamental bottleneck. Vectorisation and parallelisation reduce the constant factor, but sparse approximations or graph-based clustering strategies will ultimately be needed for next-generation kilometre-scale models.

One methodological limitation of the present framework is its reliance on block maxima, meaning only the single largest value per year per cell is used, and sub-annual extremes are missed. While this choice is theoretically justified within extreme value theory and aligns naturally with max-stable modelling, future work could explore peaks-over-threshold (POT) extensions in a spatial setting [1] to improve data efficiency while retaining theoretical consistency.

*New research directions..* By providing a clean Python API and CLI, **weatherisk** lowers the entry barrier for researchers who wish to apply spatial clustering of extremes to new domains (e.g., wind speed, sea-level pressure) or different datasets (e.g., ERA5, CMIP6 projections).

*Education..* The step-by-step pipeline design, the comprehensive test suite, and the documented code examples (Appendix Appendix A) make the package a practical resource for practitioners. A Jupyter notebook tutorial (`notebooks/weatherisk_quickstart.ipynb`) that walks through the full pipeline on synthetic data is included in the repository.

## 6. Conclusions

We have presented **weatherisk**, a Python package that implements the complete pipeline for spatial clustering of climate extremes via max-stable processes. The package covers the full preprocessing chain—STL detrending, annual block-maxima extraction, GEV fitting, and Fréchet transformation—followed by local composite-likelihood estimation of anisotropy parameters and regionalisation using Local Estimates Clustering (LEC) or Extremal Dependence Clustering (EDC) [4]. This consolidates a previously fragmented R/Shell workflow into a single, tested, pip-installable library.

The software has been verified at three complementary levels: per-function agreement with the historical R implementation of Contzen et al. [4] to  $\leq 10^{-10}$  relative error, end-to-end CMIP6 parity on a controlled subset with exact cluster counts, and broader structural and regression testing covering checkpointing, parallelism, and parameter integrity. The



full Figure 9 case-study application on the AWI-ESM-1-1-LR T63 grid confirms that **weatherisk** is both methodologically faithful to the reference implementation and scalable to publication-level workloads on HPC infrastructure.

Together, these checks give confidence that the results are correct and reproducible. The spatial clustering condenses a large climate field into a compact description of where extremes co-occur—a result that speaks directly to practical questions in reinsurance, infrastructure planning, and climate services. Two limits of the current design should be stated plainly: the block-maxima approach uses only one value per grid cell per year, so shorter-duration extremes are not captured; and the quadratic cost of pairwise comparison makes kilometre-scale grids impractical with the current algorithm.

Several directions follow naturally from these boundaries. Statistically, peaks-over-threshold extensions in a spatial setting [1] would improve data efficiency while preserving the theoretical framework, and sparse or graph-based dissimilarity approximations would address the quadratic scaling. On the engineering side, a compiled Rust extension (**weatherisk\_core**) built via PyO3 bindings [17] demonstrated a  $2.8\times$  end-to-end speed-up on synthetic benchmarks, with the local MLE step as the principal bottleneck; releasing it as a tested, opt-in backend is the most immediate next step. Looking further ahead, GPU parallelisation via CuPy [15] or Numba [16] offers a natural path to handling next-generation kilometre-scale Earth system simulations without changing the underlying statistical framework.

*Outlook: Digital Earth System Twins..* High-resolution digital replicas of the Earth system, such as those produced by the European Destination Earth initiative, require automated analysis tools operating at the same scale and cadence. Spatial clustering of extremes fits naturally into such a pipeline, converting per-cell dependence estimates into a compact, monitorable set of homogeneous regions that can be passed to impact models. Future extensions would require incremental cluster updating, linkage to socio-economic exposure databases, and objective criteria for detecting structural regime shifts.

**weatherisk** is available under the MIT licence.

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*Author contributions.* W. Kchikach designed and implemented the software, conducted all numerical experiments, and wrote the first draft of the manuscript. T. Dickhaus and G. Lohmann supervised the work, contributed to the scientific conception, and revised the manuscript. All authors read and approved the final version.

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## Appendix A. Key code examples

### Appendix A.1. A.1 Synthetic validation (Flow 1)

```
from weatherisk.parameters import get_preset
from weatherisk.simulation import simulate_non_stat
from weatherisk.grid import make_grid
from weatherisk.estimation import (
    calc_local_estimates,
    smooth_estimates,
    calc_estimates_in_clusters,
)
from weatherisk.clustering import (
    ellipse_dissimilarity,
    cluster_from_dissimilarity,
    saunders_clustering,
)
from weatherisk.plotting import plot_cluster_map

# 1. Load a parameter preset
p = get_preset("stripes")

# 2. Build a grid and simulate data
grid = make_grid(resolution=10)
data = simulate_non_stat(
    p.params, grid, n_sim=20,
    df=p.df, alpha=p.alpha, seed=42,
)

# 3. Estimate local anisotropy parameters
raw_est = calc_local_estimates(
```

```

    data, grid, df=p.df, alpha=p.alpha,
)
est = smooth_estimates(raw_est, grid)

# 4. Cluster by ellipse-overlap dissimilarity
D = ellipse_dissimilarity(est, grid)
clusters = cluster_from_dissimilarity(
    D, quantile_threshold=0.3,
)

# 5. Re-estimate within clusters
in_clust = calc_estimates_in_clusters(
    data, clusters, grid,
    df=p.df, alpha=p.alpha,
)

# 6. Plot
fig = plot_cluster_map(clusters, grid)
fig.savefig("lec_clusters.pdf")

```

Listing 3: Complete synthetic validation workflow.

#### *Appendix A.2. A.2 CMIP6 Figure 9 case-study workflow*

```

# On HPC (albedo SLURM cluster):
$ sbatch hpc/run_fig9_clean.slurm

# Or locally (requires CMIP6 data):
$ python scripts/reproduce_fig9.py \
    --data-dir /path/to/AWI-ESM-1-1-LR/pr/ \
    --workers 8

# Programmatic access:
from weatherisk.cmip6_pipeline import (
    run_cmip6_pipeline,
    CMIP6Config,
)
cfg = CMIP6Config() # paper defaults
results = run_cmip6_pipeline(cfg)
# results contains: labels_lec, labels_edc,
# k_lec, k_edc, smoothed, frechet, ...

```

Listing 4: Running the Figure 9 case study on HPC or locally.

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